



2.6 Optimisation of models (HL)

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Differentiation is used to find the maximum or minimum of a model $f(x)$ by solving $f'(x)=0$ for stationary points, then classifying each using a sign change of $f'(x)$ or the second derivative test ($f''(x)<0$ for a maximum, $f''(x)>0$ for a minimum). In applied problems (e.g. maximizing volume/area/profit or minimizing cost/material), a constraint is often used to reduce the problem to a function of one variable before optimising, and solutions must be checked against the valid domain.

Worked example: follow the reasoning

An open-top box is made from a square sheet of card of side 30 cm by cutting squares of side x cm from each corner. The volume is $V(x)=x(30-2x)^2$ cm³, $0<x<15$. Find the value of x that maximizes the volume, and the maximum volume.

Solution: $V(x)=900x-120x^2+4x^3$. $V'(x)=900-240x+12x^2=0 \rightarrow x^2-20x+75=0 \rightarrow (x-5)(x-15)=0$, so $x=5$ or $x=15$ ($x=15$ invalid, giving $V=0$). $V''(x)=24x-240$; at $x=5$, $V''=-120<0$, confirming a maximum. $V(5)=5 \times 20^2=2000$ cm³.

Quick check · multiple choice

Given $f'(x_0)=0$, which condition confirms x_0 is a local maximum using the second derivative test?

- A $f''(x_0)>0$
- B $f''(x_0)<0$
- C $f''(x_0)=0$
- D $f(x_0)=0$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A packaging cost model (in dollars) is $C(x)=x^3-40x^2+500x$ for the number of units x , $0 \leq x \leq 30$. (a) Find $C'(x)$ and hence the values of x at the stationary points. (b) Use the second derivative to classify each stationary point. (c) Find the local minimum cost within the domain, to the nearest dollar, and the corresponding value of x . [8]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $C'(x) = 3x^2 - 80x + 500 = 0$; discriminant $= 6400 - 6000 = 400$, so $x = (80 \pm 20)/6$, giving $x = 10$ or $x \approx 16.67$. (b) $C''(x) = 6x - 80$. At $x = 10$, $C'' = -20 < 0$ (local maximum); at $x \approx 16.67$, $C'' = 20 > 0$ (local minimum). (c) $C(16.67) \approx 16.67^3 - 40(16.67)^2 + 500(16.67) \approx 4629.6 - 11111.2 + 8333.5 \approx \1852 at $x \approx 16.7$ units.

Quick check answer

B. A negative second derivative at a stationary point confirms it is a local maximum (the gradient is decreasing through that point).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.